



M. Paola Vaccaro

PhD candidate in Astrophysics
Heidelberg University, Germany

✉ mariapaolavaccaro@gmail.com

🌐 mariapaolavaccaro.github.io

📄 ORCID 0000-0003-3776-9246

📖 [ADS Publication List](#)

Research interests

- Intermediate-mass black holes
- Dynamics in AGN disks
- Dynamics in star clusters
- Gravitational-wave sources
- Stellar collisions
- Einstein Telescope science

Methods and software

- **Modelling:** semi-analytical frameworks, population synthesis, SPH hydrodynamical simulations, Bayesian statistics
- **Languages:** Python, C/C++, CUDA, Fortran
- **HPC:** MPI, Slurm, GPU and cluster workflows (EuroHPC summer school, RIKEN; SSC research-software fellow)
- **Code:** lead developer of [FastClusterAGN](#) and [ARISE](#)

Languages

English (fluent, C1), Italian (native), German (basic), Spanish (basic)

Research profile

I am a computational astrophysicist and PhD student (Heidelberg University, supervisor: Prof. Michela Mapelli) modelling how black holes grow from stellar to intermediate and supermassive masses, and what their gravitational-wave mergers reveal about the early Universe. I showed that AGN disks can grow black holes through the pair-instability gap into the intermediate-mass range, but only in low-viscosity disks with efficient gas hardening, and that the resulting populations match systems like GW190521 and GW231123. Expected defence January 2028; available for postdoc and fellowship positions starting from late 2027 or early 2028.

Selected highlights

- **10 papers**, of which 4 as first author. ~ 80 citations on the first-author papers. h-index 7 (NASA ADS, Sept. 2026)
- **12 conference contributions** (EAS, MODEST, ET Symposium, among others) and seminars at ISTA in Vienna, GSSI in L'Aquila, the University of Arizona and UNAB in Santiago.
- **Invited** to write a contribution for a Royal Society theme issue (2025) and to give talks at the Niels Bohr Institute (2024) and the Physics of Data Workshop, Venice (2023).
- **Selected** for the SSC Fellowship (2026–27) in research software engineering, Scientific Software Center, Heidelberg; PhD fellow of IMPRS; member of the STRUCTURES Young Research Convent.
- **Mentoring and service:** co-supervised 3 students (1 MSc, 2 BSc); teaching assistant, 2 semesters; referee for A&A, JCAP and PRD; member of the Einstein Telescope collaboration (OSB Div. 3).

Scientific publications *as main author*

- **Black hole populations and hierarchical growth across the AGN parameter space**, [M. P. Vaccaro](#), M. Mapelli, A. A. Trani, B. Liu. [arXiv:2606.10823](#), accepted for publication in A&A (2026)
- **The role of migration traps in the formation of binary black holes in AGN disks**, [M. P. Vaccaro](#), Y. Seif, M. Mapelli. [A&A, 708, A171 \(2026\)](#)
- **Astrophysics: a modern discipline with a Newtonian origin**, [M. P. Vaccaro](#). [Phil. Trans. R. Soc. A, 383: 20230294 \(2025\)](#), invited contribution to a theme issue
- **Impact of gas hardening on hierarchical black hole mergers in AGN disks**, [M. P. Vaccaro](#), M. Mapelli, C. Perigois, et al. [A&A, 685, A51 \(2024\)](#)

Education

Sep. 2023 – 2028 **PhD in Astronomy and Cosmic Physics**

At Institute for Theoretical Astrophysics (ITA), ZAH, Heidelberg University in Heidelberg, Germany.

Supervisor: Prof. Michela Mapelli - Thesis title: “Binary Black Hole population in the Local Universe”

Oct. 2020 – Sep. 2022 **Master degree in Astrophysics**

At Università degli Studi di Padova in Padua, Italy. Final grade: 110/110 *cum laude*, GPA: 29.4/30.

Supervisor: Prof. Michela Mapelli - Thesis title: “Hierarchical Black Holes Mergers in Active Galactic Nuclei”

Sep. 2019 – Jul. 2020 **Master courses in Matter Physics**

At Università di Pisa in Pisa, Italy - GPA: 28.8/30

Sep. 2016 – Sep. 2019 **Bachelor degree in Physics**

At Università di Pisa in Pisa, Italy. Final grade: 110/110 *cum laude*, GPA: 28.6/30. Supervisor: Prof.

Pier Giorgio Prada Moroni - Thesis title: “Cooling evolution of white dwarfs and cosmochronology”

Sep. 2011 – Jul. 2016 **High school diploma**

At Liceo Scientifico Alfano da Termoli in Termoli (CB), Italy. Final grade: 100/100 *cum laude*

Aug. 2014 – Jul. 2015 **High school exchange year**

At Mead High School in Mead, Colorado, USA. GPA: 4.0/4 - ACT score: 33/36

Fellowships, funding and grants

PhD in Astronomy and Cosmic Physics (2023 – 2028, [expected](#)) Heidelberg University. Funded by the European Research Council (ERC Consolidator Grant DEMOBLACK, no. 770017, PI: M. Mapelli) and by the German Excellence Strategy via the Heidelberg Cluster of Excellence STRUCTURES (EXC 2181 – 390900948).

SSC Fellowship (2026 – 2027) Selected for individual training in research software engineering and sustainable development at the [Scientific Software Center](#), Heidelberg University. Competitive selection; one of **5** awardees.

Pre-doctoral fellowship ([February 1st – August 31st, 2023](#)) in the DEMOBLACK ERC Team at Università degli Studi di Padova, Padua, Italy. Supervisor: Prof. Michela Mapelli.

Internship ([October 1st – December 31st, 2022](#)) in the GW group at IFAE-UAB, Barcelona, Spain. Supervisors: Dr. Lluïsa-Maria Mir i Martínez and Prof. Mario Martínez Pérez.

Travel grants: Research visit to the GSSI in L’Aquila, Italy, awarded through the ACME Transnational Access programme ([April 2026](#)); participation in the IAU Symposium 398 and MODEST-25 hosted in Seoul, South Korea, sponsored by the International Astronomical Union ([July 2025](#)); participation in the International HPC Summer School hosted in Kobe, Japan, sponsored by the EuroHPC Joint Undertaking via the HPC SPECTRA project ([July 2024](#)).

Memberships

Einstein Telescope collaboration ([October 1st, 2023 – present](#)), Observational Science Board (OSB), Division 3.

IMPRS ([October 1st, 2023 – present](#)) PhD fellow of the 19th generation in the International Max Planck Research School (IMPRS) for Astronomy and Cosmic Physics at Heidelberg University.

Young Research Convent ([October 1st, 2023 – present](#)) of the Heidelberg Cluster of Excellence STRUCTURES. Programme funded by the German Excellence Strategy.

Scientific publications

as co-author

- **When black holes align: a subpopulation of aligned massive BBHs observed via GWs**, Stefano Rinaldi, Charmaine Wong, [M. P. Vaccaro](#), et al., 2026, arXiv:2606.23305
- **The Science of the Einstein Telescope**, Adrian Abac et al. (incl. [M. P. Vaccaro](#)), 2025, arXiv:2503.12263
- **The boring history of Gaia BH3 from isolated binary evolution**, Giuliano Iorio et al. (incl. [M. P. Vaccaro](#)), A&A, 690, A144 (2024)
- **Hierarchical binary black hole mergers in globular clusters: mass function and evolution with redshift**, Stefano Torniamenti et al. (incl. [M. P. Vaccaro](#)), A&A, 688, A148 (2024)
- **GW190521: a binary black hole merger inside an active galactic nucleus?**, Sophia Morton et al. (incl. [M. P. Vaccaro](#)), Phys. Rev. D, 108, 123039 (2023)
- **The hitchhiker's guide to the galaxy catalog approach for gravitational wave cosmology**, Jonathan R. Gair et al. (incl. [M. P. Vaccaro](#)), AJ, 166, 22 (2022)

Conferences and workshops

- **Contributed Talk | Many faces of stellar-mass black holes** Kathmandu, Nepal
Talk title: "AGN Disks as Multi-Messenger Factories for Spinning Binary Black Holes". **October 11th–16th, 2026**
- **Poster Presentation | Gamma Ray Symposium** | 31 August–4 September, 2026 Heidelberg, Germany
- **Flash Talk + Poster Presentation | EAS annual meeting** [SwissTech Convention Centre](#), Lausanne, Switzerland
Session: "SS31: Exotic populations in star clusters seen through the lens of binary interactions". **29 June–3 July, 2026**
Poster title: "Multi-Stage Collision Products in Dense Clusters".
- **Contributed Talk | XVI Einstein Telescope Symposium** [RWTH Aachen University](#), Aachen, Germany
Talk title: "Cosmic Evolution of Binary Black Hole Mergers from AGN Disks". **June 15th–19th, 2026**
- **Contributed Talk | MODEST-25 & IAU Symposium 398** [Seoul National University](#), Seoul, South Korea
Talk title: "Hierarchical Black Hole Mergers in AGN Disks". **June 16th–20th, 2025**
- **Contributed Talk | Dynamix** [Institute of Astronomy](#), Cambridge, UK
Title: "Hierarchical BH Mergers in AGN Disks: Tracing MBH Growth Across Cosmic Time". **June 2nd–6th, 2025**
- **Invited Talk | New ideas on the origin of black hole mergers** [Niels Bohr Institute](#), Copenhagen, Denmark
Talk title: "Specks of light from AGNs as possible indicators of the dynamical formation of binary black holes". **August 12th–16th, 2024**
- **Contributed Talk | EAS annual meeting** [Congress Center](#), Padova, Italy
Session: "S1: Unveiling black hole Growth across Cosmic Time in the JWST and LISA era". **July 1st–5th, 2024**
Talk title: "Hierarchical Black Hole Mergers in Active Galactic Nuclei Discs".
- **Contributed Talk | Massive Black Holes in the First Billion Years** [University College Cork](#), Cork, Ireland
Talk title: "Possible constraints on the SMBH mass distribution from hierarchical BBH mergers in AGN discs". **April 29th – May 3rd, 2024**
- **Contributed Talk | RESCEU-NBIA workshop on GW sources** [University of Tokyo](#), Tokyo, Japan
Talk title: "The impact of Gas Hardening on Hierarchical Black Hole Mergers in Active Galactic Nuclei Disks". Contribution to the [discussion panel](#) "Many-Body Systems". **December 7th–15th, 2023**
- **Flash Talk | Workshop "GwPopNext"** [University of Milano Bicocca](#), Milan, Italy
Talk title: "Semi-analytical Models for Hierarchical Binary Black Hole Mergers". **July 10th–14th, 2023**
- **Poster Presentation | Renaissance of Stellar Black-Hole Detections** [Lorentz Center](#), Leiden, Netherlands
Invited participation **June 26th–30th, 2023**
- **Invited Talk | Physics of Data Workshop** [Venetian institute of science, literature and art](#), Venice, Italy
Talk title: "How do black holes grow? Active galactic nuclei as a breeding ground for intermediate-mass black holes" **April 19th–21st, 2023**

Colloquia

- Galaxy Group Meeting (online) UNAB, Santiago, Chile | August 28th, 2026
- Department Colloquium GSSI, L'Aquila, Italy | April 24th, 2026
- Institute Seminar ISTA, Vienna, Austria | November 13th, 2025
- “BinBoom” Group Meeting (online) The University of Arizona | September 30th, 2025
- Physics of Stellar Objects Group Meeting HITS, Heidelberg, Germany | April 16th, 2025
- Gravitation and Cosmology Group Meeting GSSI, L'Aquila, Italy | December 9th, 2024
- ITP Cosmology Seminar ITP, Heidelberg, Germany | October 29th, 2024
- Einstein Telescope monthly meeting (online) ET-OSB Div3 | October 25th, 2024
- “Astroscopicini” Seminar (online) GSSI, L'Aquila, Italy | April 26th, 2024
- Galaxy Group Meeting ITA-ZAH, Heidelberg, Germany | March 27th, 2024
- Planet Formation Group Meeting ITA-ZAH, Heidelberg, Germany | February 20th, 2024

Summer Schools attendance

- “Cosmic Multiplicity: Evolution and Fate of Binary Stars” Summer School September 14th-18th, 2026
hosted by IMPRS in Heidelberg, Germany
- “AI for Astronomy” Summer School hosted by IMPRS in Heidelberg, Germany September 8th-12th, 2025
- International HPC Summer School hosted by the RIKEN Center in Kobe, Japan July 7th-12th, 2024
- “ML4ASTRO, Machine Learning in Astrophysics” workshop hosted by HITS in Heidelberg June 2024; July 2025
- GW Open Data Workshop hosted by GWOSC (online event) May 23rd-25th, 2022

Teaching experience and academic service

- Reviewer for *Astronomy & Astrophysics (A&A)*, *Journal of Cosmology and Astroparticle Physics (JCAP)*, *Physical Review D (PRD)* for a total of 5 papers. since 2025
- Teaching assistant for “Introduction to Astronomy and Astrophysics”: Summer Semesters 2024, 2025
Conducted daily exercise tutorials, corrected student submissions, and provided detailed feedback.
- Student co-supervision
 - Master student Julian L. Saling, “Hydrodynamical simulations of massive-star collisions” 2025/2026
 - Bachelor student Yannick Seif, “Pair Up of Binary Black Holes in Active Galactic Nuclei” 2024/2025
 - Bachelor student Jona A. Schubert, “The mass ratio of binary black holes: impact of mass transfer” 2024
- Volunteer staff member at EAS Annual Meetings July 2024, July 2026

Outreach and Media Coverage

- Research featured in *Gist.Science*: “The formation of binary black holes in AGN disks”. April 8th, 2026
- Author of “The Origins of Black Hole Mergers: Diverse Pathways Across the Universe”, March 7th, 2025
blog post for the STRUCTURES Cluster of Excellence, aimed at a general audience.
- Co-organizer for the *workshop* “A dance of stars and black holes” of the Girls’ Day program 2024, 2025, 2026
“Physik hautnah!” for school-aged students
- Member of the organizing committee for the *Scholé* “School of Physics” for high schoolers since 2022